

JEE MAIN REVISION MASTER: LOGARITHM

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 8 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Logarithm - Core Application	8 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Logarithm - Core Application (High Yield — 8 PYQs)

Core Principle: Key problem pattern in Logarithm.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Let $\alpha=(1/4)+(1/8)+(1/16)+\dots\infty$ and $\beta=(1/3)+(1/9)+(1/27)+\dots\infty$. Then the value ..." [Ans: C]
- [2026 Evening Shift] Q#2 (MCQ): "The sum of all the real solutions of the equation $\log_{((x+3))(6x^2+28x+30)}=5-2\log_{((6x+10))(x^2+6x+9)}$ is equal to :..." [Ans: N/A]
- [2025 Morning Shift] Q#3 (MCQ): "The product of all solutions of the equation $e^{5(\log_e x)^2+3}=x^8, x>0$, is :..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 4th April Evening Shift (Q#1)

Let $\alpha = (1/4) + (1/8) + (1/16) + \dots$ and $\beta = (1/3) + (1/9) + (1/27) + \dots$. Then the value of $(0.2)^{\log_{(\sqrt{5})}(\alpha)} + (0.04)^{\log_{(5)}(\beta)}$ is equal to :

- (A) 4
- (B) 5
- (C) 8 **[Correct]**
- (D) 25

Correct Answer: C

Solution Analysis: We first find the values of α and β . Given $\alpha = (1/4) + (1/8) + (1/16) + \dots$. This is an infinite geometric progression with $a = (1/4)$, $r = (1/2)$. So, $\alpha = (a / (1-r)) = ((1/4) / (1-(1/2))) = ((1/4) / (1/2)) = (1/2)$. Now, $\beta = (1/3) + (1/9) + (1/27) + \dots$. This is also an infinite geometric progression with $a = (1/3)$, $r = (1/3)$. Thus, $\beta = (a / (1-r)) = ((1/3) / (1-(1/3))) = ((1/3) / (2/3)) = (1/2)$. So both are equal to $(1/2)$. Now the expression becomes